FĂTU MIRUNA MARIA
311 CA
Miruna

Studiul Efectului HALL

TABEL DE VALORI

B(mT)		-300	-250	-200	-150	-100	-50	0	50	100	150	200	250	300
I (mA)														
10	U_H (mV)	-48	-45	-43	-40	-37	-34	-30	-28	-25	-22	-20	-18	-15
20		-77	-74	-69	-64	-58	-51	-44	-38	-32	-28	-23	-17	-13

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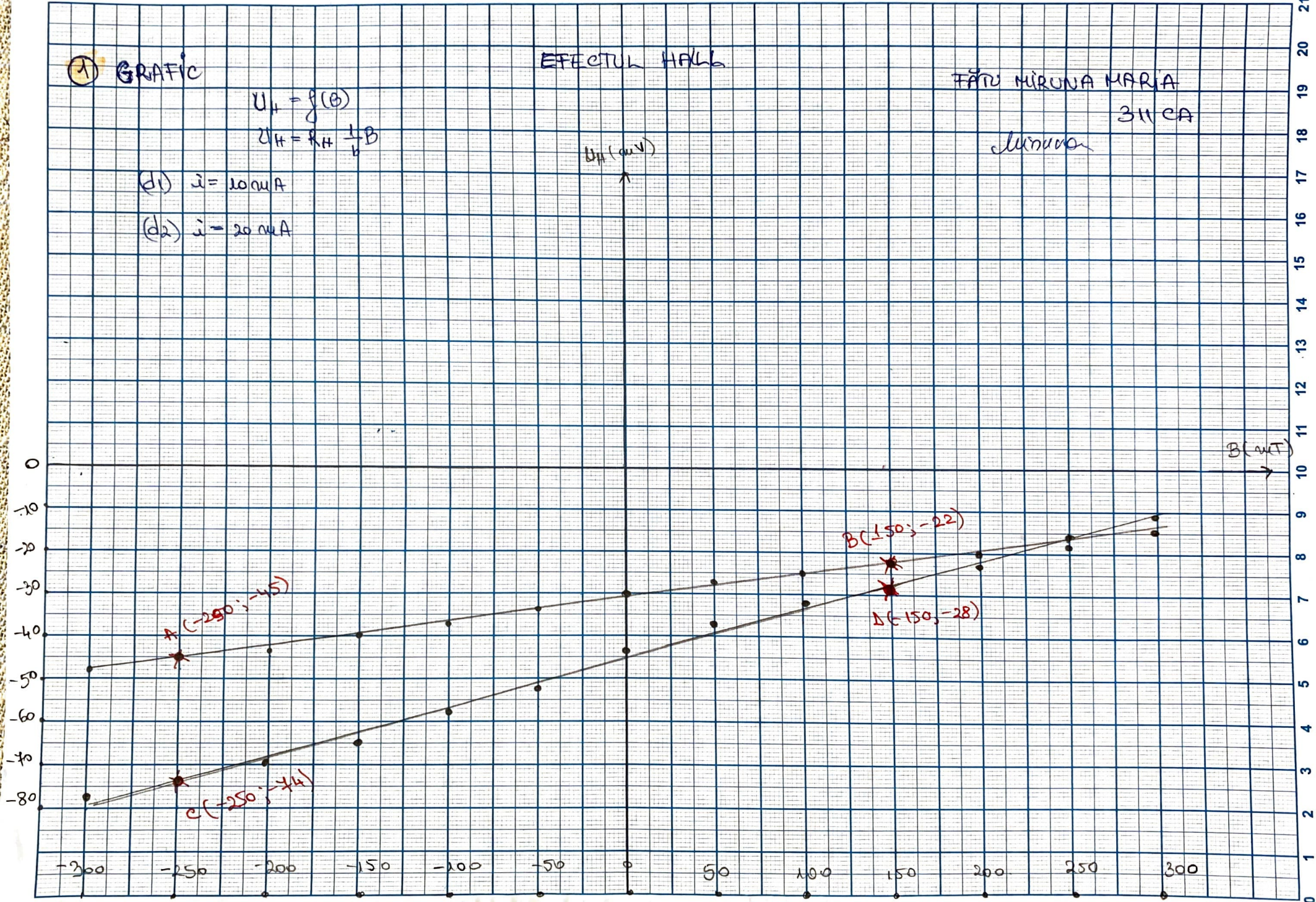
1) GRAFIC

$U_H = f(B)$

$U_H = R_H \frac{1}{b} B$

(d1) $i = 10$ mA

(d2) $i = 20$ mA



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② PANTA DREPTELOR:

$$U_H = f(B), \; U_H = R_H \frac{I}{b} B \Rightarrow m = R_H \frac{I}{b}$$
$$= mx + n$$

1 pt $I = 10\,mA$

$$m_1 = \left(\frac{\Delta y}{\Delta x}\right) = \frac{\Delta U_H}{\Delta B} = \frac{U_{HB} - U_{HA}}{B_B - B_A}$$

$$m_1 = \frac{(-22 - (-45)) \cdot 10^{-3} V}{(150 - (-250)) \cdot 10^{-3} T} = \frac{23}{400} \frac{V}{T} = 0,0575 \frac{V}{T}$$

2 pt. $I = 20\,mA$

$$m_2 = \left(\frac{\Delta y}{\Delta x}\right) = \frac{\Delta U_H}{\Delta B} = \frac{U_{HD} - U_{HC}}{B_D - B_A}$$

$$m_2 = \frac{(-28 - (-74)) 10^{-3} V}{(150 - (-250)) \cdot 10^{-3} T} = \frac{46}{400} \frac{V}{T} = 0.115 \frac{V}{T}$$

③ CONSTANTA HALL:

$$\left.\begin{aligned} U_H &= R_H \frac{I}{b} B \\ U_H &= f(B) = mx + n \end{aligned}\right| \Rightarrow m = R_H \frac{I}{b} \Rightarrow R_H = \frac{bm}{I}$$

Caz 1: $I = 10\,mA$, $b = 1\,mm$ $\Rightarrow R_H = \frac{10^{-3} \cdot 0,0575}{10 \cdot 10^{-3}} \frac{mV}{T \cdot A} = 0,00575 \cdot \frac{m \cdot J/C}{\frac{Kg}{A s^2} \cdot A} = m \cdot \frac{m^2}{s^2} \cdot \frac{kg}{C} \cdot \frac{s^2}{kg}$

$\Rightarrow R_H = 0,00575\; m^2/C$

Caz 2: $I = 20\,mA$, $b = 1\,mm$ $\Rightarrow R_H = \frac{10^{-3} \cdot 0,115}{20 \cdot 10^{-3}} \cdot \frac{m^3}{C} = 0,00575 \frac{m^3}{C}$

$\Rightarrow R_H = 0,00575 \frac{m^2}{C}$

(*) MEDIA

(4) CONCENTRAȚIA GOLURILOR

$R_H = \frac{1}{pe} \Rightarrow p = \frac{1}{eR_H}$

$\overline{R_H} = \frac{R_{H1} + R_{H2}}{2} \Rightarrow p = \frac{1}{R_H} = \frac{1}{0,00565 \cdot 1,6 \cdot 10^{-19}}$

$= \frac{10^{19}}{0,00904}$

$p = \frac{1}{1,6 \cdot 10^{-19} \cdot 0,00575} = \frac{10^{19}}{0,0092} = 108,6 \cdot 10^{19} \cdot m^{-3}$

(concentrația golurilor)

(5) → conductivitate

$\sigma = \frac{1}{\rho}$ → rezistivitate

$a = 10 \text{ mm}$

$b = 1 \text{ mm}$

$c = 20 \text{ mm}$

$R = 35 \Omega$

$\rho = R \cdot \frac{A}{e} \Rightarrow \rho = R \cdot \frac{a \cdot b}{c}$

$\Rightarrow \sigma = \frac{c}{R \cdot a \cdot b}$

$\sigma = \frac{20 \cdot 10^{-3}}{35 \cdot 10 \cdot 10^{-3} \cdot 10^{-3}} = 0,05714 \cdot 10^{-3} \frac{1}{\Omega m} = 57,14 \frac{1}{\Omega m}$

(6) mobilitatea golurilor

$\mu_p = \sigma R_H$

$\mu_p = 57,14 \cdot 0,00575 = 3,28 \frac{1}{\Omega m} \cdot \frac{m^3}{C} = 0,328 \frac{A}{V} \cdot \frac{m^2}{As} = 0,32 \cdot \frac{m^2}{Vs}$

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